

Report of Project 7: Transported Composition PDF Method

Russ, Rachel

*Department of Aeronautics and Astronautics
Purdue University
April 23, 2026*

1 Lifted Turbulent Jet Flame in a Vitiated Co-Flow

1.1 Grid Information – Q3(a)

The solver used for the experiment is very computationally expensive, so to keep the computational time requirements within reason, the calculations are performed on a very coarse grid. This makes the results not as accurate as they could be if the mesh was finer. Also, the grid is not fine enough to resolve the boundary layer, so the fuel tube wall is treated as a symmetry boundary.

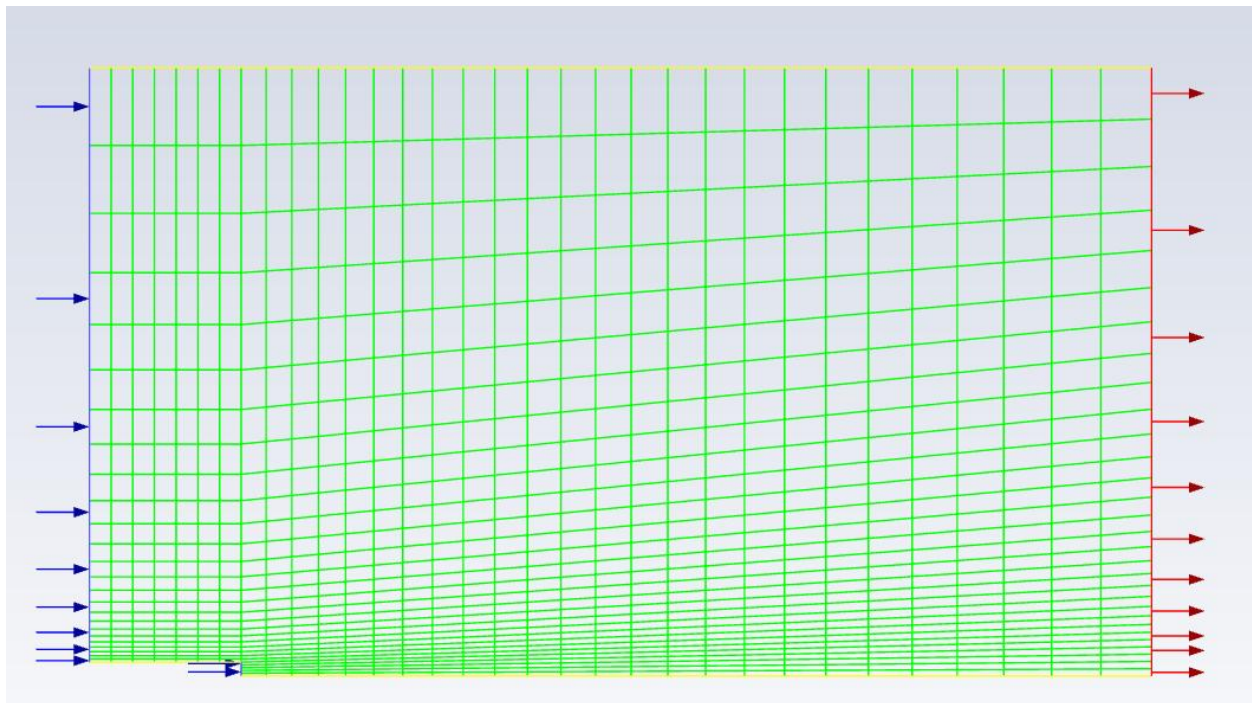


Figure 1 Mesh of FLUENT Calculation

Fig. 1 shows the mesh that is used for the calculations. There are 925 cells, 1912 faces, and 988 nodes total in the mesh.

Table 1 Different Mesh Surfaces and Locations

Surface	Location and Purpose of Surface
Axis	Axis of symmetry of axisymmetric problem (bottom yellow line in Fig. 1)
Coldflow	Oxidizer inlet (large left side surface with blue arrows in Fig. 1)
Jet	Fuel inlet (small indented left surface with dark blue arrows in Fig. 1)
Outlet 1	Bottom most outlet, primarily mixed (bottom right-side surface with red arrows in Fig. 1)
Outlet 2	Top outlet, primarily oxidizer and much larger than Outlet 1 (top right-side surface with red arrows in Fig. 1)
Top 1	Top edge of tube, top left surface and much smaller than Top 2, primarily unburned oxidizer (top left yellow line in Fig. 1)
Top 2	Top right edge of tube and includes mixed fuel and oxidizer (top right yellow line in Fig. 1)
Tube	Edge of flame tube, treated as a symmetry boundary for this problem and is where the fuel enters the mixing chamber (offset bottom left yellow line in Fig. 1)

Table 1 shows the location and purpose of each of the surfaces in the test case. The interior surface is not mentioned in this table as it is where the mixing occurs and its location and purpose is clear.

1.2 Cold Flow Test Case – Q4(c)

When volumetric reactions are turned off in the solver, it turns the case into a cold flow. There is turbulence accounted for. Cold cases are important as they show if a model is stable, accurate, and computationally efficient.

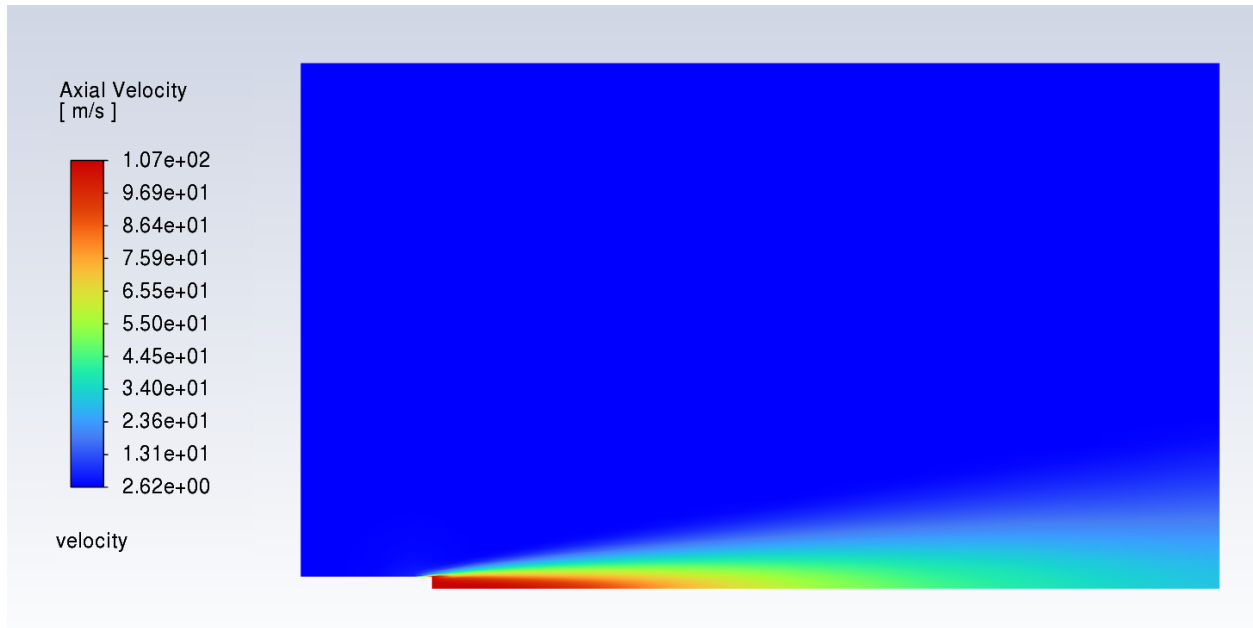


Figure 2 Mean Axial Velocity Contour Plot

Fig. 2 shows the contour of the axial velocity of the cold case. The mean axial velocity is very similar to the axial velocity as the contour is already very smooth. The mean values allow for a smoother contour, but that is not needed here. It can be shown that the velocity is greatest near the fuel inlet and over a horizontal distance gets diffused to a lower velocity. This is the expected trend.

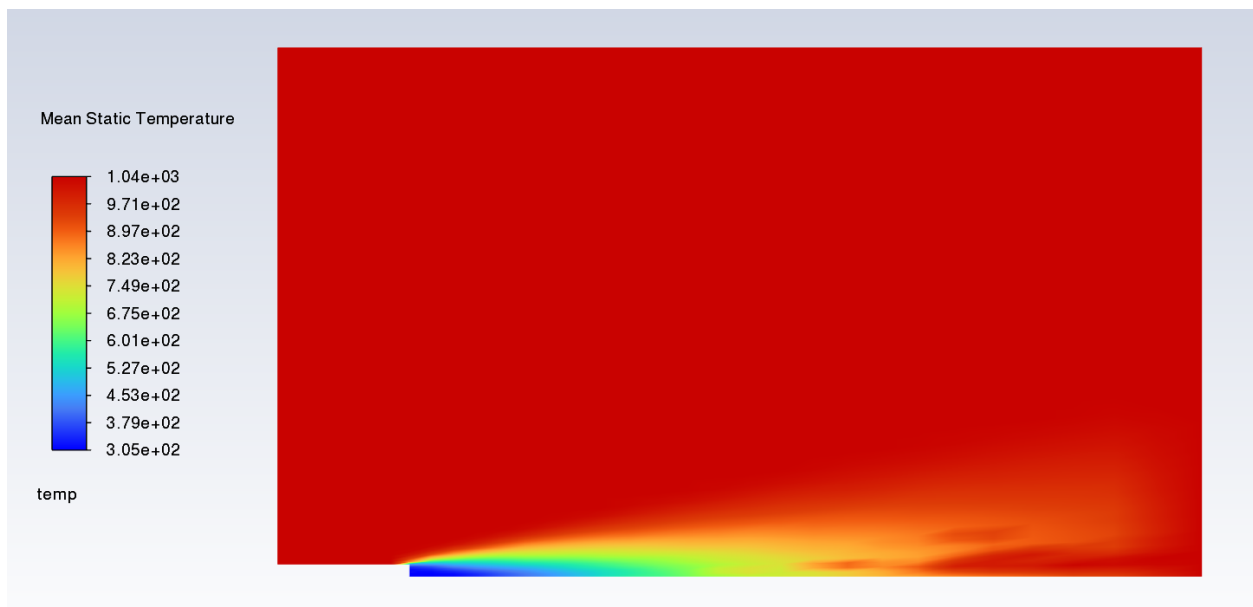


Figure 3 Mean Static Temperature Contour Plot

Fig. 3 shows the contour of the mean static temperature. Because of the course grid, the temperature contour is not that smooth, so the mean temperature is displayed to make the graph slightly smoother. The temperature is greater for the oxidizer than the fuel, which is because the fuel is set to 305 K while the oxidizer is set to 1045 K for the initial conditions. Because this is a cold flow case, the fuel does not ignite which is why the temperature does not increase. As the fuel and oxidizer combine, their temperatures go towards an equilibrium state which cools the oxidizer and heats the fuel as the horizontal distance increases. The contour shows the expected behavior.



Figure 4 Mean OH Mass Fraction Contour Plot

Fig. 4 shows the contour of the mean OH mass fraction. The mass fraction of OH is zero throughout the entire test area. This is due to the fact the case does not ignite. Because it does not ignite, there are no products of combustion, which OH is. This shows that the contour is what is expected.

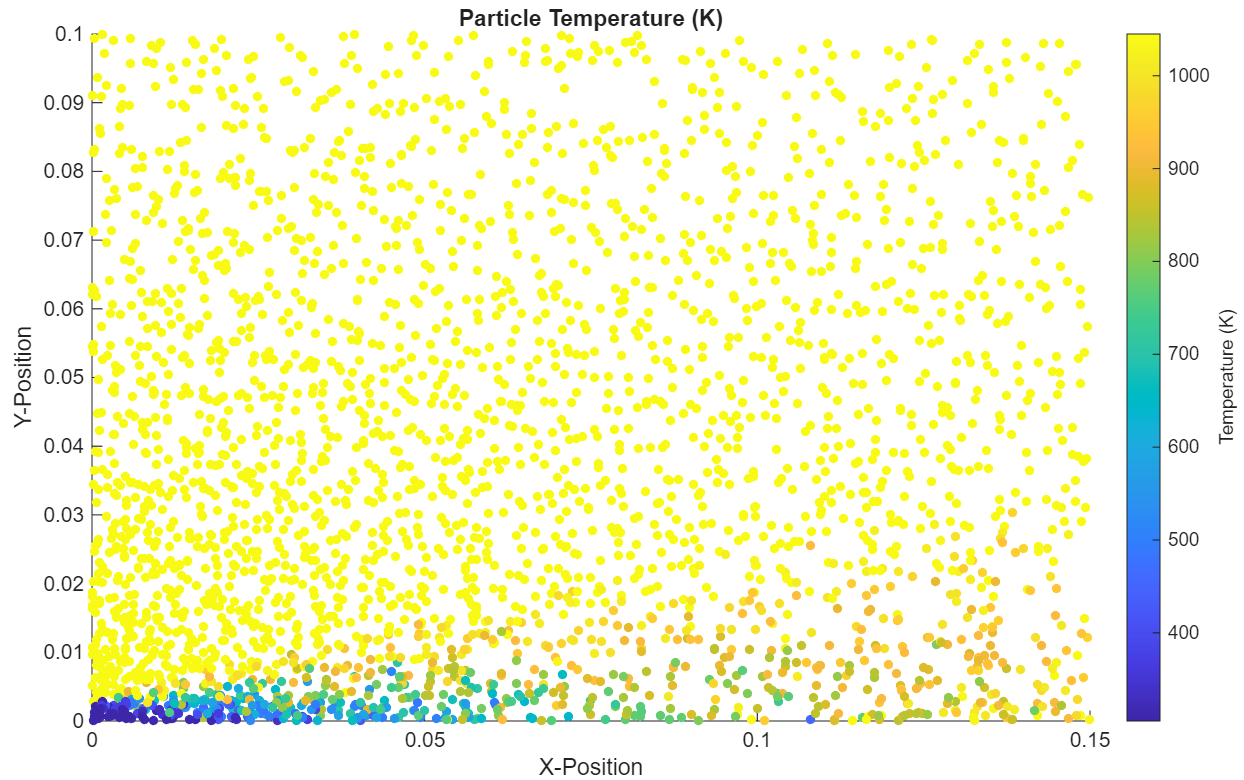


Figure 5 Particle Temperature

Fig. 5 shows the particle temperature plotted against the x and y positions. Fluent has a bug which makes it not plot the particle temperature, but the output data is not bugged so the data is processed in MATLAB instead which means it cannot be plotted against the grid. This data has a skip of 2 (every other particle is skipped) incorporated in it to make it less dense and easier to see. It makes sense that the oxidizer is a larger temperature than the fuel as the inlet temperature of the oxidizer is larger, and the temperature diffuses as the particles mix which makes sense and matches the contour graph of the mean temperature (Fig. 3) which is to be expected.

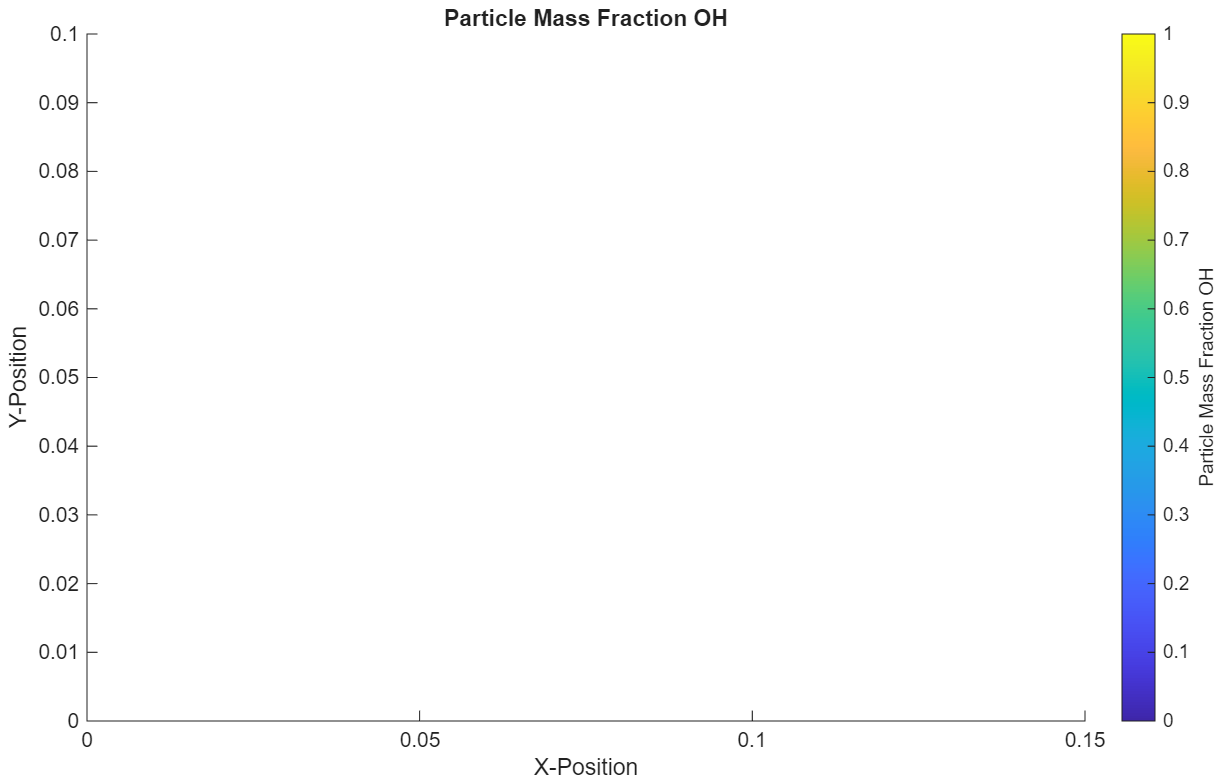


Figure 6 Particle Mass Fraction of OH

Fig. 6 shows the particle mass fraction of OH. There are no particles present because the case is a cold flow case where no fuel is burned. This means that there is no OH produced. This particle graph is consistent with what is expected.

1.3 Hot Flow Test Case – Q5

The volumetric reactions are turned on to make this case a hot fire case. A stiff chemistry solver is used. The flame is lifted with the base of the flame around 5 mm of the horizontal axis. The 0 x-coordinate indicates the x-distance where the fuel inlet is, so the negative x-coordinate values are where only the oxidizer is at a horizontal distance before the fuel is let into the system.

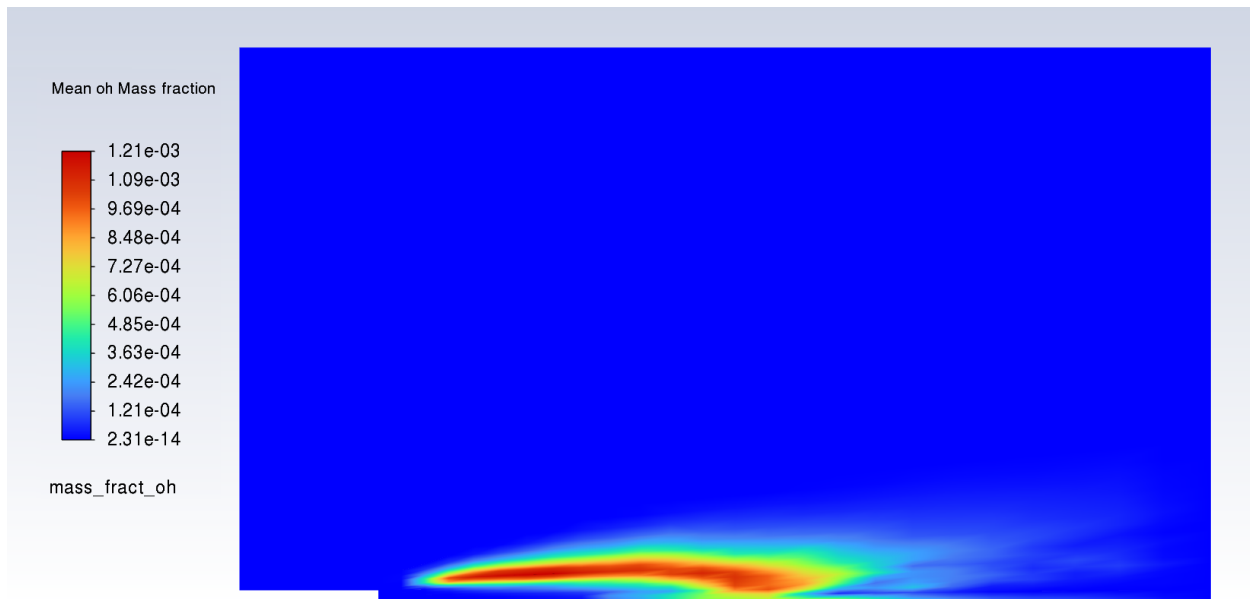


Figure 7 Contour of Mean Mass Fraction of OH

Fig. 7 shows the contour of the mean mass fraction of OH. The mass fraction of OH shows where the flame is as OH is a byproduct of combustion, and it is a radical, so it does not exist out of the physical flame. Due to this, it is shown that the flame is lifted off the fuel inlet, and it is hollow due to the lack of OH near the axis of the flame. It is slightly closer to the inlet than expected; however, the residuals are converged. Other than that, this is the contour that was expected.

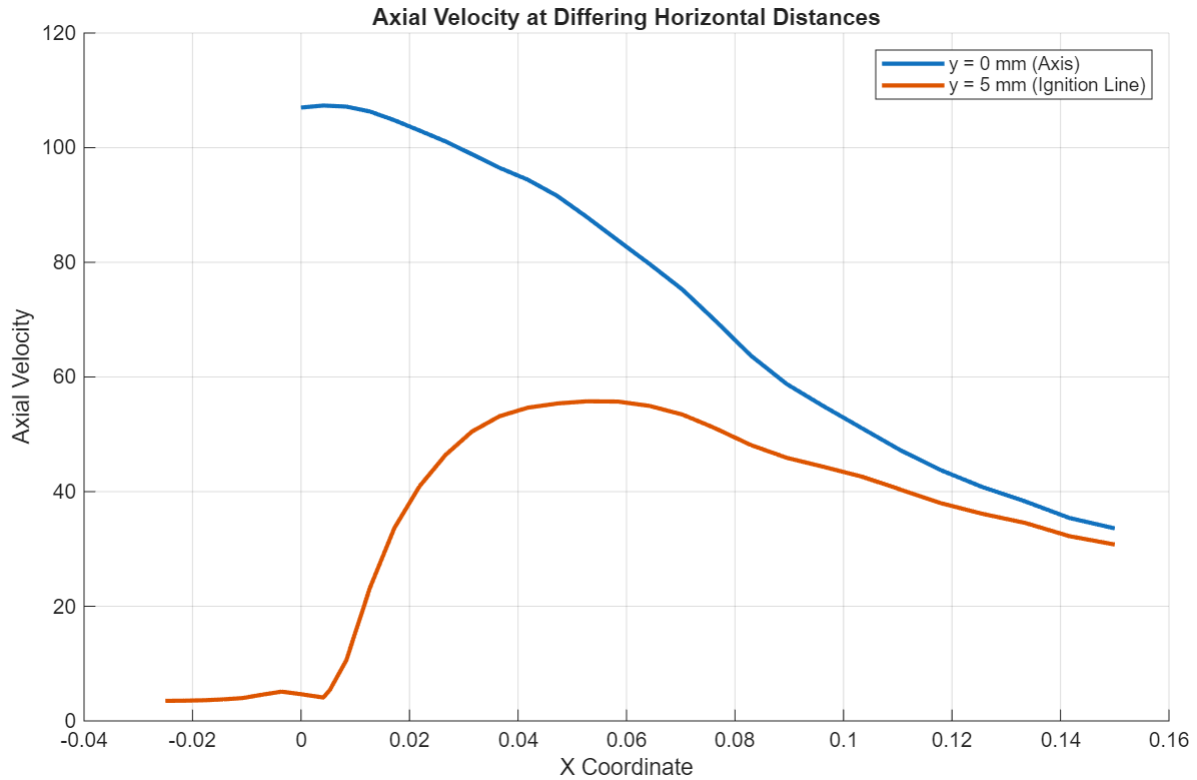


Figure 8 Mean Axial Velocity at the Axis and Ignition Line at Differing Horizontal Locations

Fig. 8 shows the mean axial velocity at differing horizontal locations at two locations in the y-direction. It can be shown that the axial velocity is consistently lower at the ignition line than at the axis. This is because the initial velocity of the fuel is much higher than the oxidizer, and the oxidizer is inserted at a higher y-distance. That data follows the same trend of velocity decreasing as x-coordinate increases after the flame has ignited. However, the ignition line shows where the flame is better as there is a peak at that location (0.05 m).

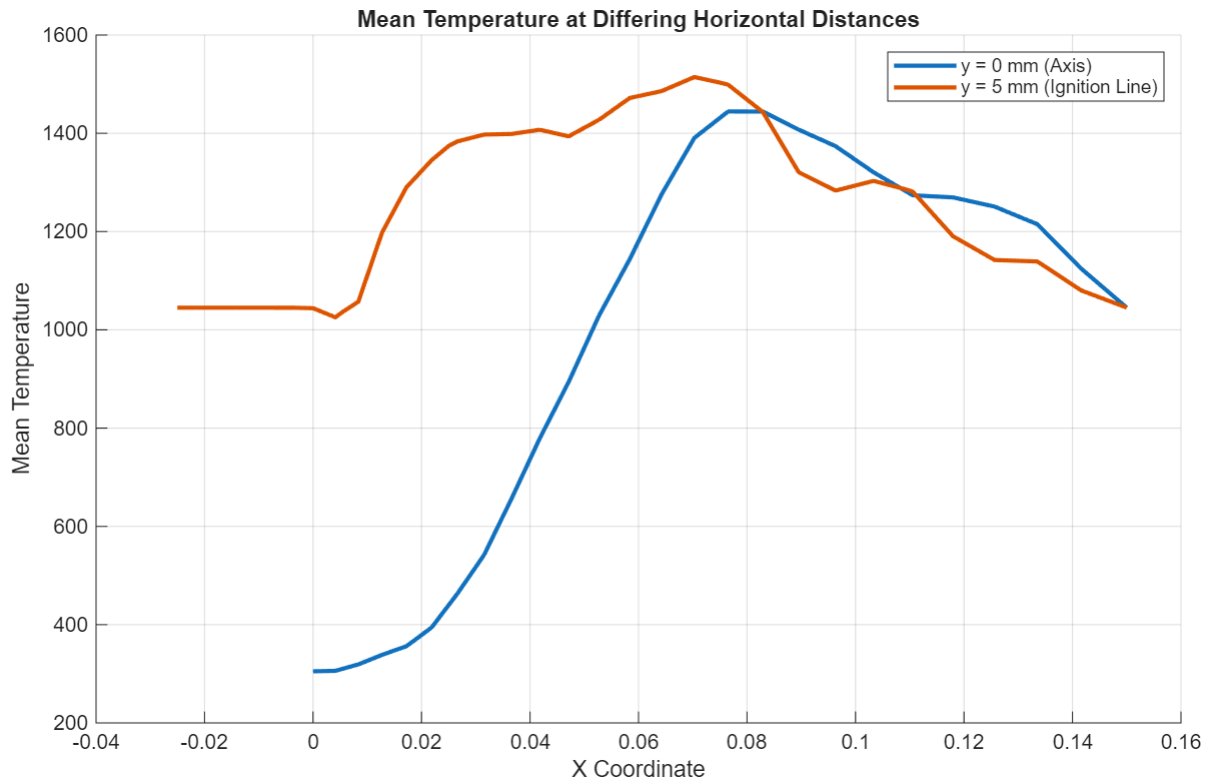


Figure 9 Mean Temperature at the Axis and Ignition Line at Differing Horizontal Locations

Fig. 9 shows the mean temperature at differing horizontal locations at two locations in the y-direction. The mean temperature is much higher at the ignition line than at the axis for the first half of the flame tube, then the axis and ignition line have relatively the same temperature which decreases as horizontal distance increases. The initial temperature of the oxidizer is greater than the initial temperature of the fuel which is why the ignition line has a much higher temperature initially. The ignition line is also where the flame starts, and at the base of the flame, the flame is hallowed as there is no flame at the axis while there is at the ignition line. This explains why the temperature of the ignition line is higher than the axis line at the base of the flame.

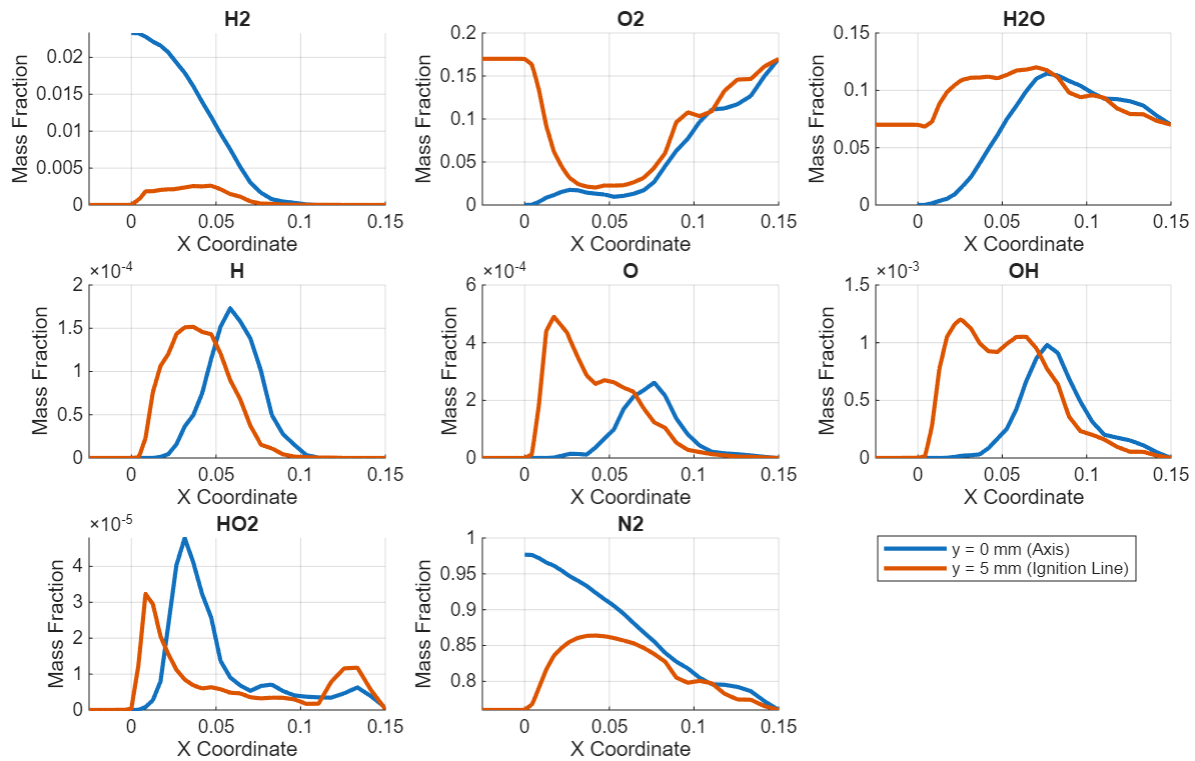


Figure 10 Mass Fractions at the Axis and Ignition Line at Differing Horizontal Locations

Fig. 10 shows the mass fractions at differing horizontal locations at two locations in the y-direction. The residuals (OH, O, H, and HO_2) help show where the flame is as they get consumed very fast. The OH graph shows that the flame occurs at the ignition line much sooner than the axis which makes sense with the contour graph (Fig. 7) and the mean temperature graph (Fig. 9). The OH graph also shows that the end of the flame (at a greater horizontal distance) is roughly the same at each y-location (axis and ignition line). The other residuals show this trend as well which confirms this theory. The O_2 is at its lowest mass fraction where the flame is as it is getting burnt up. The amount of O_2 increases at the end of the flame due to the excess of it available around the flame, and for the most part the mass fraction is the same for the axis and ignition line. The same thing in reverse occurs with the H_2 and N_2 as both molecules are present in the fuel. The mass fraction of H_2 and N_2 are much greater at the axis than the ignition line as the flame occurs at the ignition line, so the fuel gets consumed. The amount of H_2 and N_2 decreases as the flame gets weaker as there is less fuel to be burned. The H_2O molecule does a good job of showing where the flame is as well as it is a stable product of the flame. There is a greater amount of H_2O on the ignition line as this is where the flame is thus there is more product. These results are what is consistent with a lifted flame which shows the Fluent model does a good job of simulating it.

1.4 Mean Mixture Fraction, RMS Mixture Fraction, Turbulence Frequency, and Modelled Mean Scalar Dissipation – Q6

Because a Composition PDF Transport Model is used for the species solver to help decrease the computational expense of this case, there are multiple variables that are not calculated. Among these are the mean mixture fraction ($\tilde{\xi}$), root mean squared (RMS) mixture fraction (ξ'), turbulence frequency (ω), and modelled mean scalar dissipation ($\tilde{\chi}$). Because of this, these values need to be calculated as custom field functions.

$$\tilde{\xi} = \frac{(\tilde{Y}_{N_2} - Y_{N_2_{ox}})}{Y_{N_2_{fuel}} - Y_{N_2_{ox}}} \quad (1)$$

Eq.(1) shows the equation used for the mean mixture fraction ($\tilde{\xi}$) where $\tilde{Y}_{N_2}$ is the mean mass fraction of N_2 . $Y_{N_2_{fuel}}$ and $Y_{N_2_{ox}}$ are the mass fraction of N_2 in the fuel and oxidizer respectively. Nitrogen can be used because it is not used up in the flame and is thus linearly related to the mixture fraction. Eq.(2) and Eq. (3) show their values respectively.

$$Y_{N_2_{ox}} = 0.76 \quad (2)$$

$$Y_{N_2_{fuel}} = 0.9767 \quad (3)$$

Eq.(4) shows the equation used for the RMS mixture fraction (ξ') where Y_{N_2}' is the RMS mass fraction of N_2 .

$$\xi' = \frac{Y_{N_2}'}{|Y_{N_2_{fuel}} - Y_{N_2_{ox}}|} \quad (4)$$

$$\xi' \equiv \sqrt{\widetilde{\xi'^2}} \quad (5)$$

Eq.(5) is the definition of the RMS mixture fraction; however, Eq.(4) is the one used in practice as it has values that Fluent calculates or is inputted.

$$\omega \equiv \frac{\varepsilon}{k} \quad (6)$$

Eq.(6) shows the equation for the turbulence frequency (ω). ε is the turbulent kinetic energy, and k is the turbulent dissipation rate. These two values are calculated by Fluent.

$$\tilde{\chi} \equiv \frac{2\xi'^2\varepsilon}{k} \quad (7)$$

Eq.(7) shows the equation for the modelled mean scalar dissipation ($\tilde{\chi}$). The values used to calculate this parameter are already described for Eq.(6) and Eq.(4).

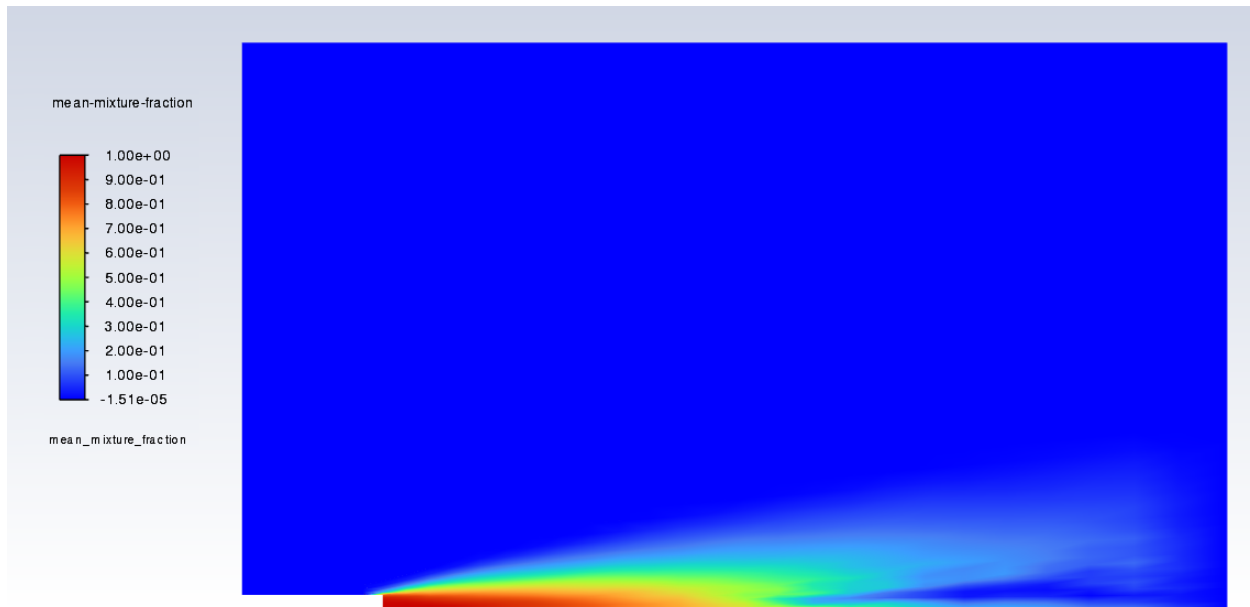


Figure 11 Contour of Mean Mixture Fraction

Fig. 11 shows the contour of the mean mixture fraction. The mixture fraction is almost zero where the oxidizer is, and it is one at the fuel inlet. This is consistent with the definition of mixture fraction as a value of one is assigned for pure fuel and a value of zero is assigned for pure oxidizer. The mixing between the two is also shown, and it is consistent with what would be expected. This proves that the assumption of Eq.(1) holds as it is a good match to the definition and it shows data that is expected.

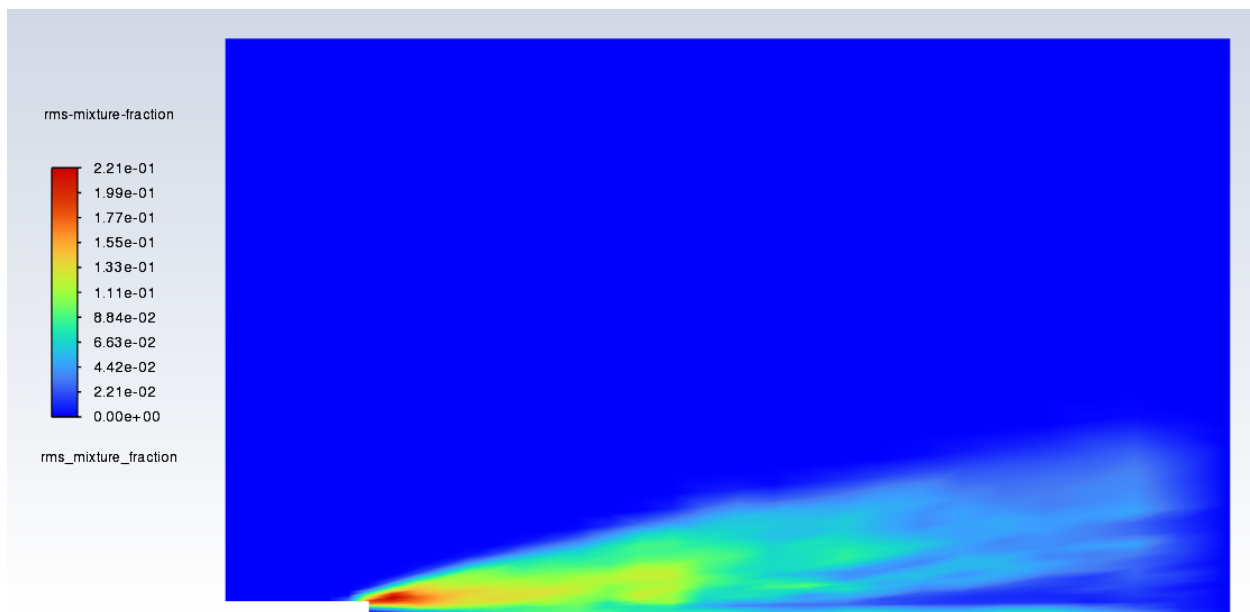


Figure 12 Contour of RMS Mixture Fraction

Fig. 12 shows the contour of the root mean squared mixture fraction. There is a much higher value near the edge of the fuel inlet, and there is a value of zero where there is pure oxidizer. This is consistent with the fact that RMS mixture fraction is the measure of variance of the fuel-oxidizer mixing in turbulent combustion. It makes sense that the most turbulent mixing occurs right at the edge of boundary of the fuel inlet which is what the contour shows. This also shows that the assumptions made in Eq.(4) hold as it is a good match to the definition and it shows data that is expected.

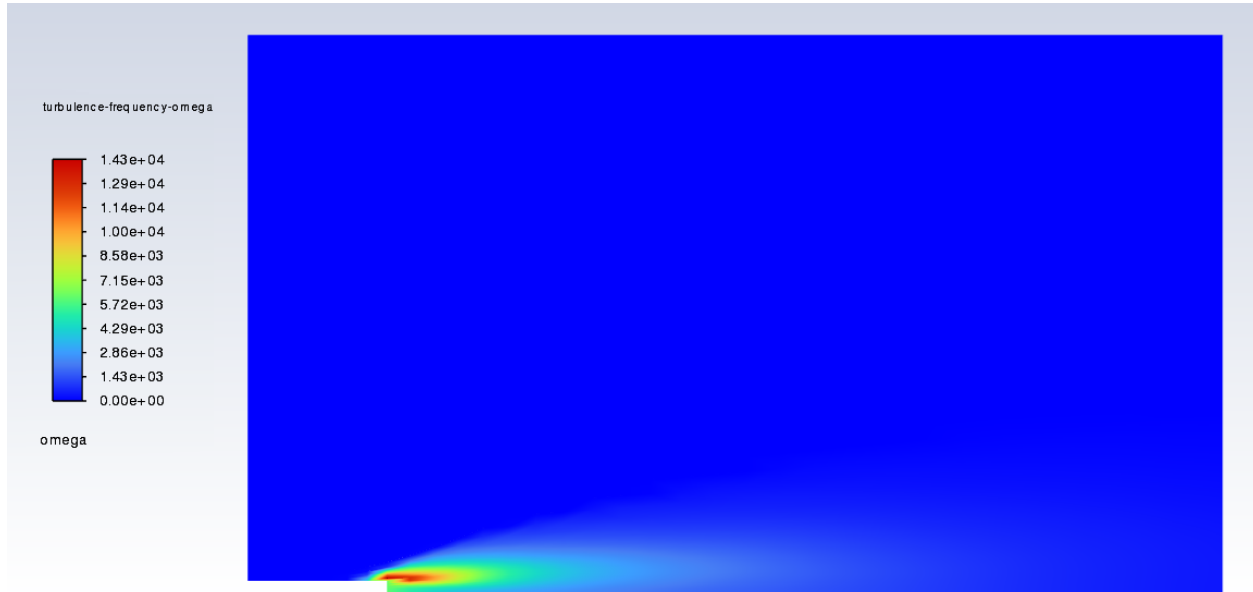


Figure 13 Contour of Turbulence Frequency

Fig. 13 shows the contour of the turbulence frequency. It shows a very large value near the edge of the fuel boundary where the fuel inlet is, and a very low value outside of this. This is consistent with what is expected as there is the most turbulence near where the flame inlet interacts with the oxidizer. Both things are traveling at vastly different speeds with a different chemical composition. Due to this, this location having the most turbulent flow is to be expected.

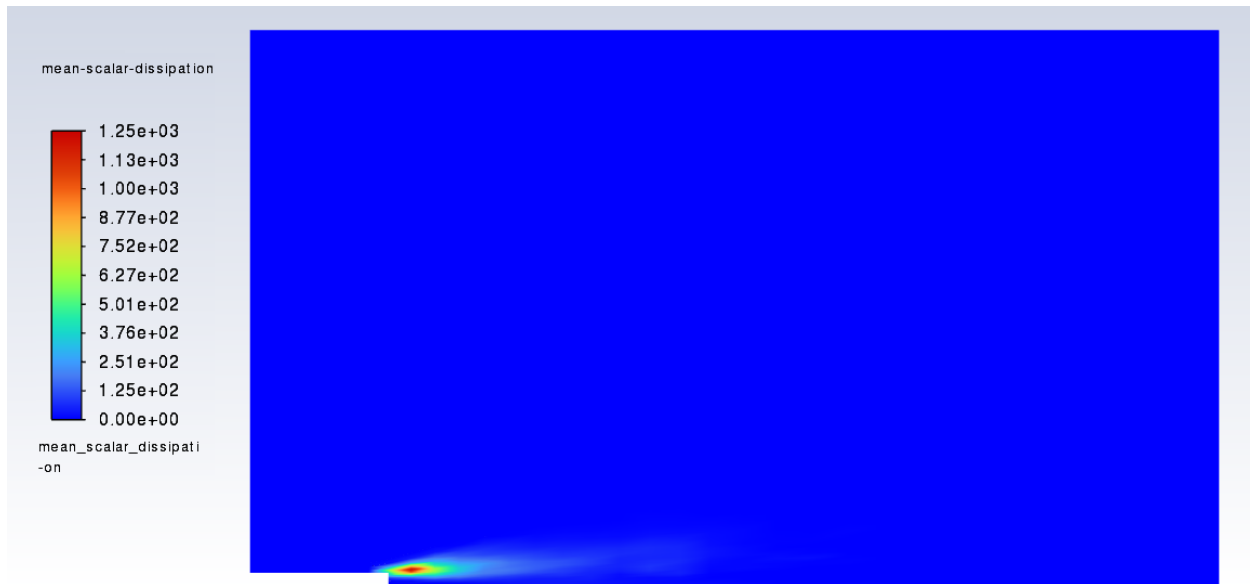


Figure 14 Contour of Mean Scalar Dissipation

Fig. 14 shows the contour of the mean scalar dissipation. The location where this occurs is very similar to the location where there is a maximum turbulent frequency. This makes sense as both values measure the turbulence, with the mean scalar dissipation measuring mixing. Due to this, it makes sense that the maximum turbulent mixing occurs at the edge of the fuel inlet where the velocities of the two differing fluids are vastly different.

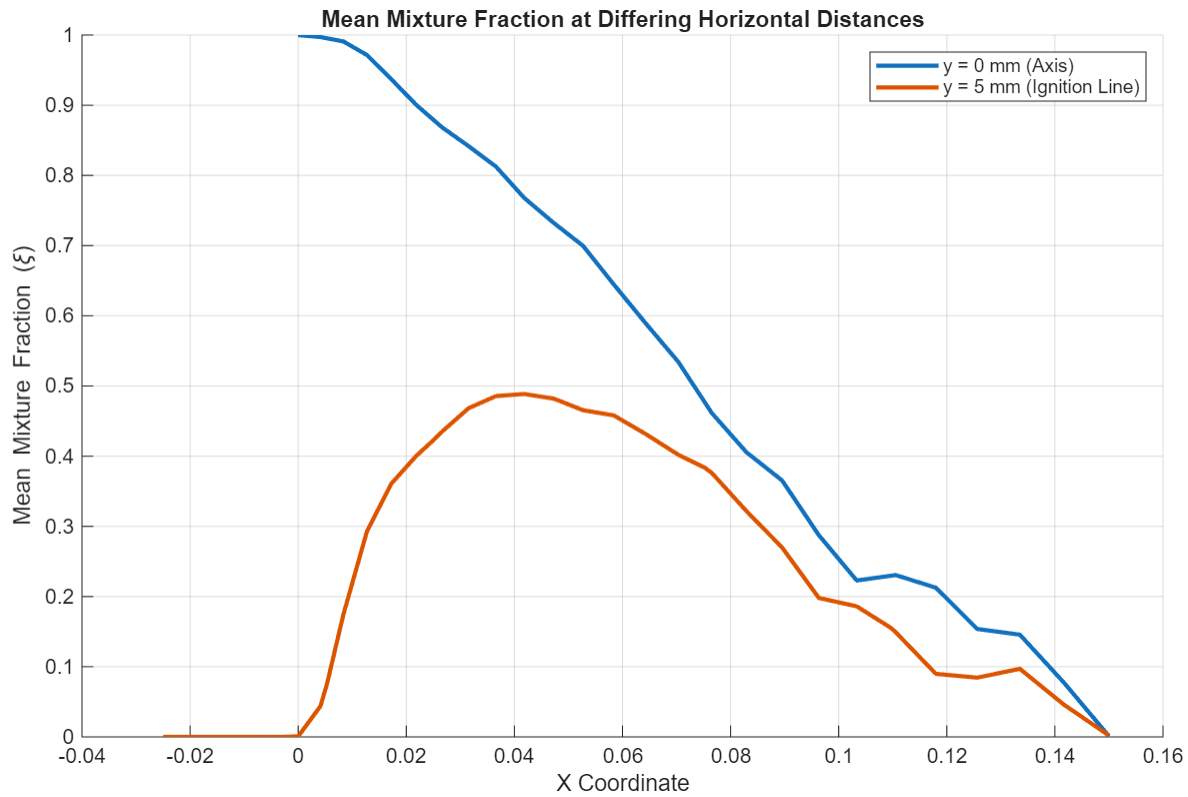


Figure 15 Mean Mixture Fraction at the Axis and Ignition Line at Differing Horizontal Locations

Fig. 15 shows the mean mixture fraction at differing horizontal locations at two locations in the y-direction. It can be shown that the mean mixture fraction is considerably higher along the axis and decreases as the horizontal distance increases. This is since the axis contains the fuel inlet, and a mixture fraction is one for pure fuel. The increase in mixture fraction across the ignition line shows where the flame occurs as the fuel is getting mixed into the oxidizer then consumed by the flame. This makes sense when compared to the contour (Fig. 11) and it is also consistent with the theory. This means the model does a good job of predicting mixture fraction.

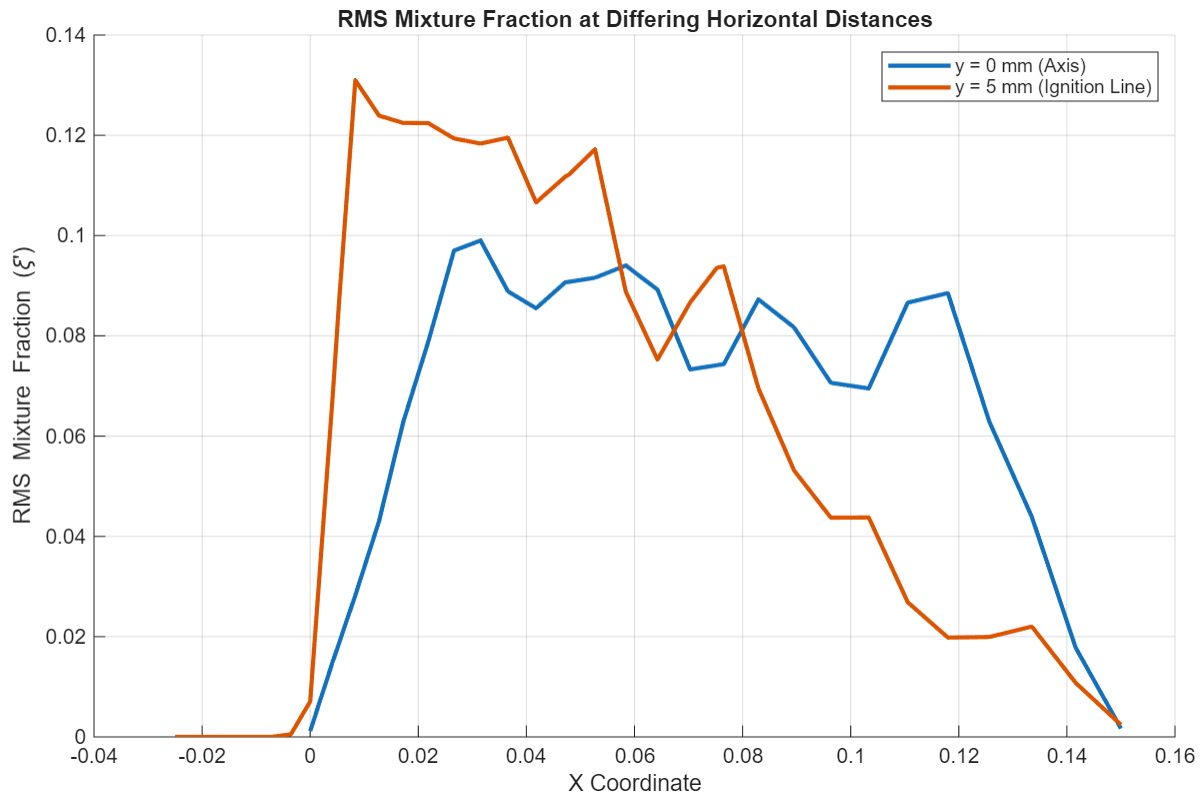


Figure 16 RMS Mixture Fraction at the Axis and Ignition Line at Differing Horizontal Locations

Fig. 16 shows the root mean squared mixture fraction at differing horizontal locations at two locations in the y-direction. The RMS mixture fraction is higher at the ignition line than for the axis near the middle of the flame tube, then it decreases rapidly. The RMS mixture fraction at both the axis and ignition line peaks around the horizontal middle of the flame tube. This is because RMS mixture fraction is the measure of variance of the fuel-oxidizer mixing in turbulent combustion. The maximum RMS value is around where the fuel and oxidizer will mix at each point. This is consistent with the definition. This also explains why the RMS mixture fraction drops at an increasing horizontal distance as there is less turbulence mixing occurring as the fuel and oxidizer are more thoroughly mixed at that point while at the inlet they are at significantly different velocities and are not mixed.

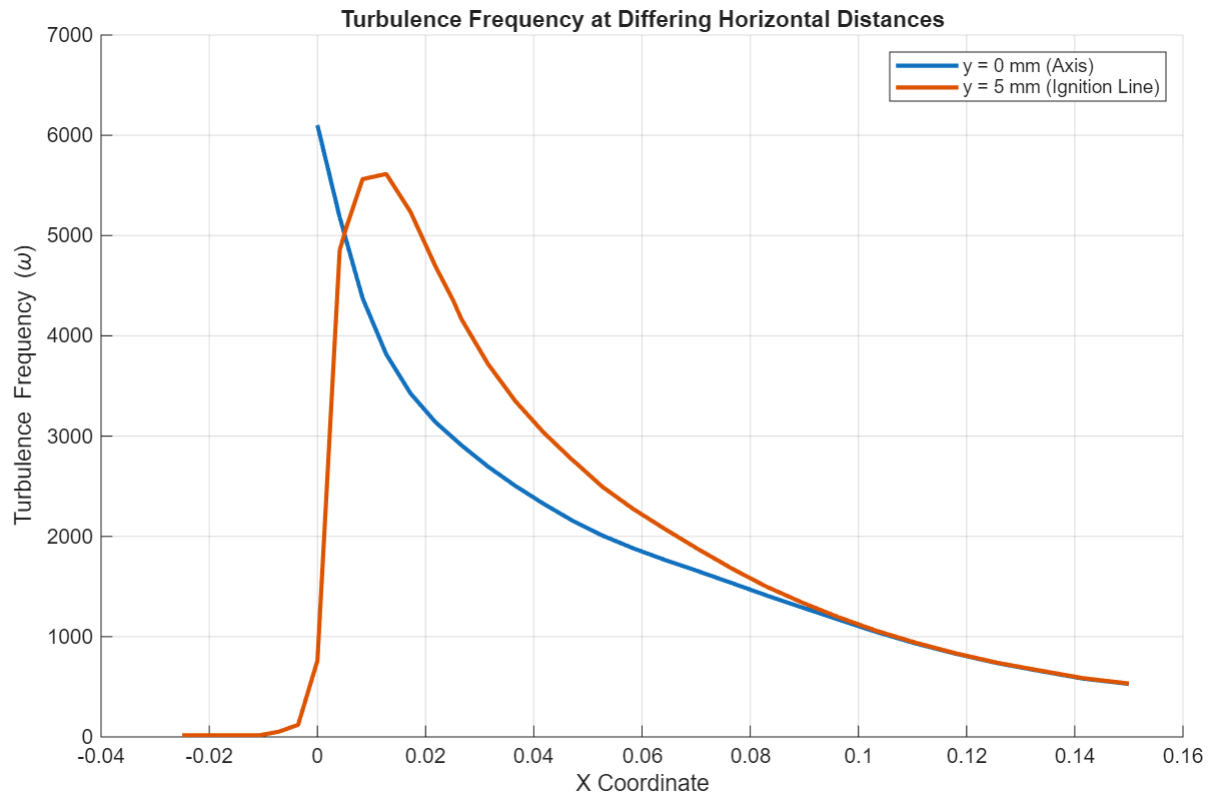


Figure 17 Turbulence Frequency at the Axis and Ignition Line at Differing Horizontal Locations

Fig. 17 shows the turbulence frequency at differing horizontal locations at two locations in the y-direction. The maximum turbulence frequency occurs at the fuel inlet and it at the axis. The maximum turbulence frequency is slightly delayed at the ignition line which makes sense as it is not where the fuel inlet is, so the fuel and oxidizer are more mixed at that point. The most turbulence frequency is where the fuel and oxidizer are first mixed due to reasons stated. This value decreases in an exponential decay which makes sense as gradually over the horizontal distance the fuel and oxidizer will mix and the flame is ignited.

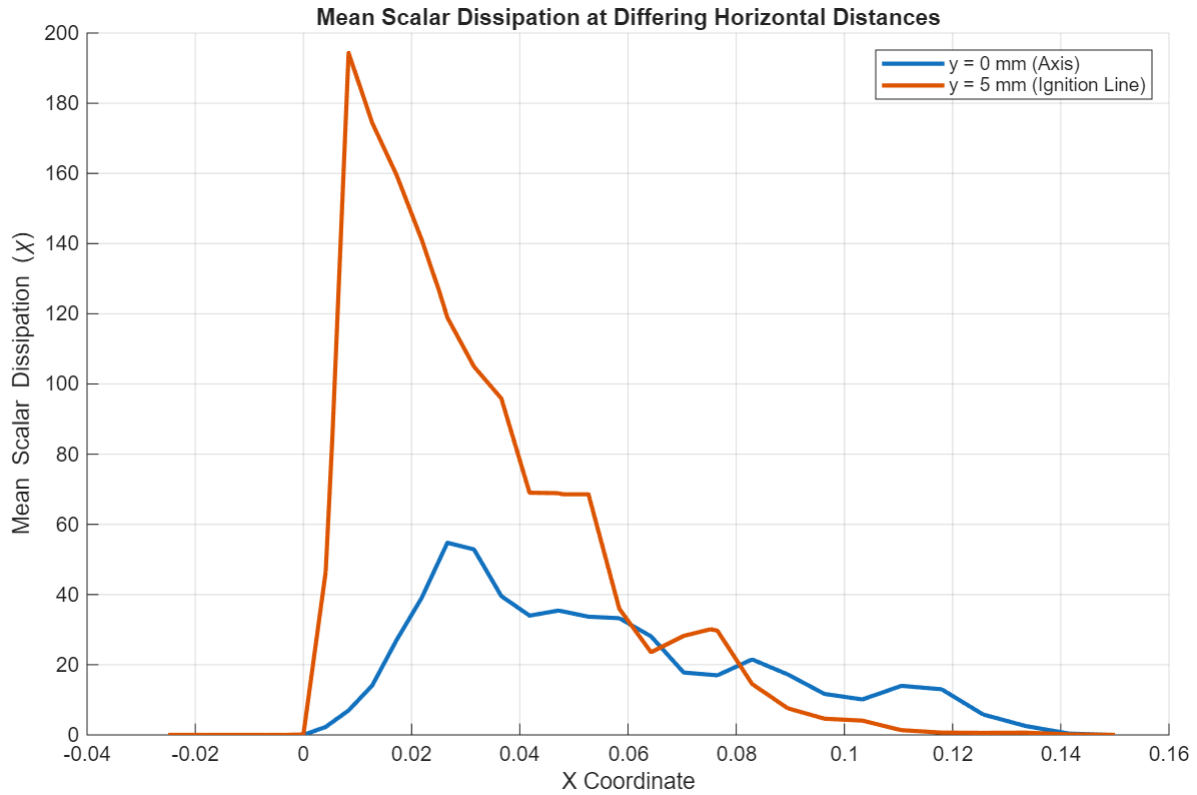


Figure 18 Mean Scalar Dissipation at the Axis and Ignition Line at Differing Horizontal Locations

Fig. 18 shows the mean scalar dissipation at differing horizontal locations at two locations in the y-direction. The mean scalar dissipation is greatest on the ignition line right after the fuel inlet. This is consistent with the fact this is where the mixing occurs the most as this is where the fuel inlet interacts with the oxidizer. The value for mean scalar dissipation decreases with increasing horizontal distance which is consistent with the fact that the fuel and oxidizer become more mixed as the horizontal distance increases. The fact that the mean scalar dissipation is lower on the axis than the ignition line also makes sense, as well as the fact that these two lines converge at increasing horizontal distance due to the mixing of the fuel and oxidizer.

1.5 Mean OH and Particle OH Comparison – Q7

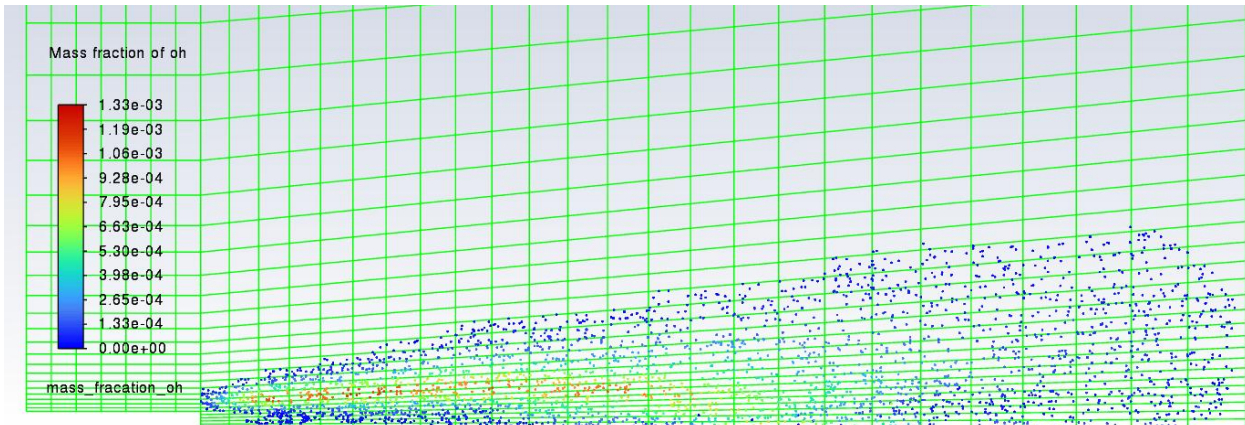


Figure 19 Mean OH Scatter Plot

Fig. 19 shows the mean OH scatter plot from Fluent. The base of the flame is zoomed in on, but a portion of the upstream region is included. The mean mass fraction of OH is not broken in Fluent so it was able to be plotted over the grid. Fluent has a bug with the particle tracking, and it would not work for the particle mass fraction of OH. The data itself was not working, which meant that MATLAB could not be used to bypass the problem. Multiple different options were explored; however, none of them worked. The particle OH scatter plot should look very similar to the mean OH scatter plot (Fig. 19). This also looks like the contour plot for the mean mass fraction of OH (Fig. 7), and this is to be expected.

1.6 Influence of Number of Particles per Cell and Iteration Increment – Q8

The default number of particles per cell is 20 for the composition PDF transport model for the model of Fluent used, and the iteration increment is set to 0.5 (originally at 0.2) to change the monitored quantities. 1000 iterations are done, which is the same amount as the other test cases.

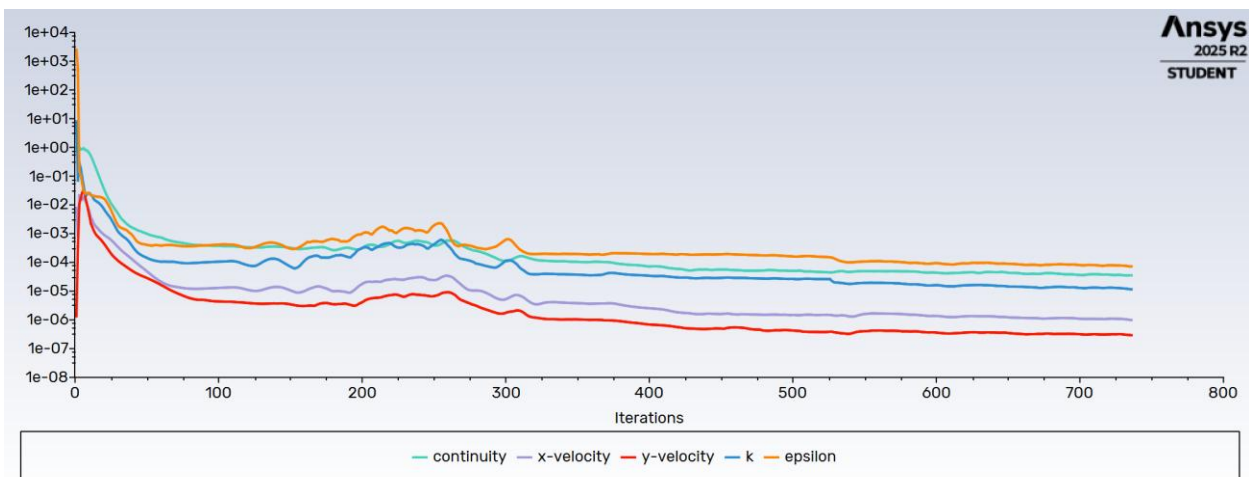


Figure 20 Residuals with Iteration Increment = 0.2 (Original)

Fig. 20 shows the scaled residuals of the base, default case. The model reaches convergence at around 350 iterations. The model gets close to convergence after around 100 iterations.

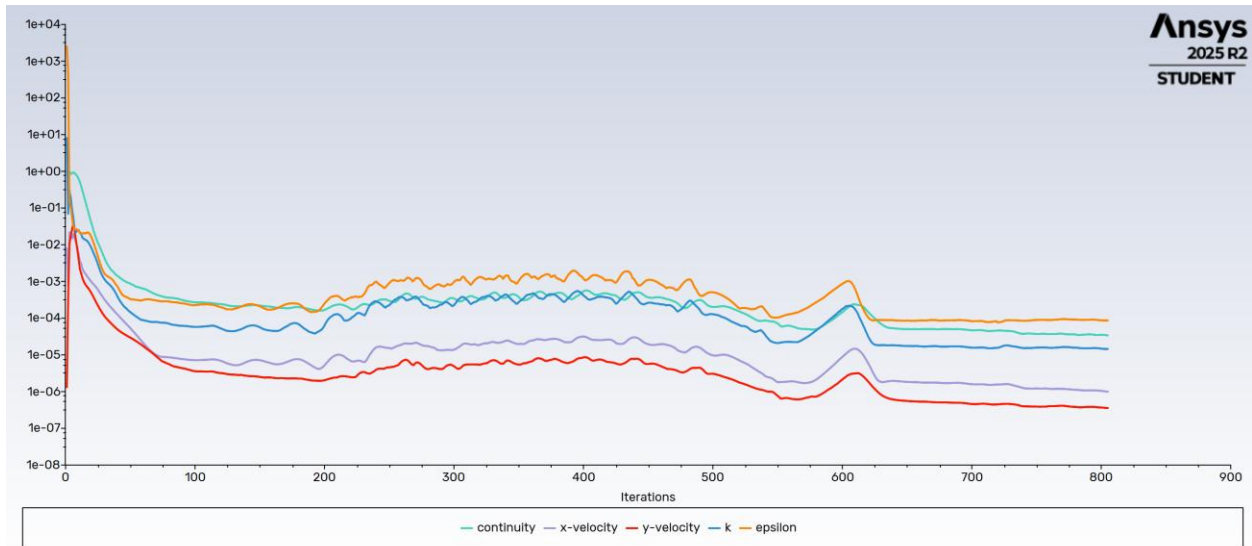


Figure 21 Residuals with Iteration Increment = 0.5 (Changed)

Fig. 21 shows the scaled residuals of the changed case. The model reaches convergence at around 650 iterations, around twice as much as the original case. The model gets close to convergence at around 100 iterations, which is the same as the default case. The converged value for each scaled residual is the same in both cases. In this model, the increase in iteration increment causes the model to not converge as fast; however, both models converge at the same values.

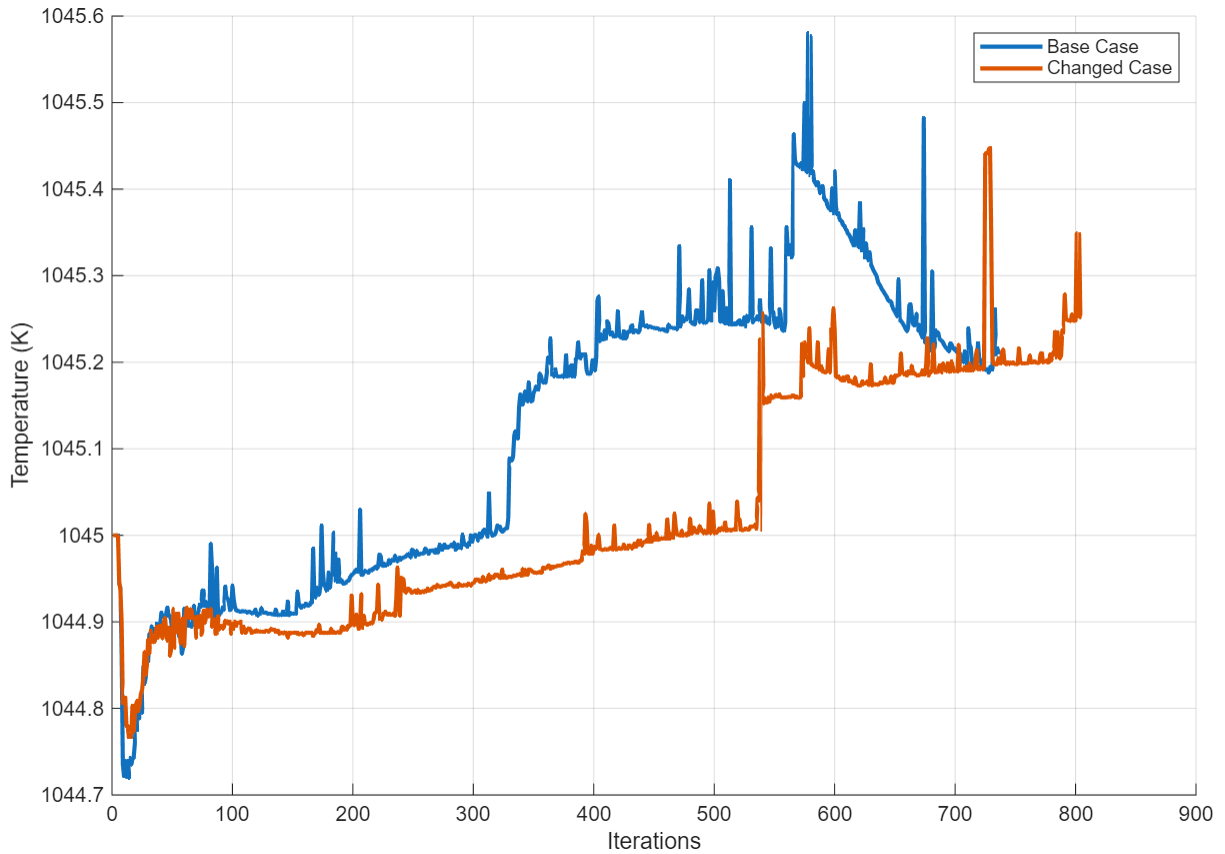


Figure 22 Vertex Maximum of Temperature at each Iteration for both Cases

Fig. 22 shows the vertex maximum of temperature (the monitored quantity) as a function of iteration. It can be shown that it seems like the base case (iteration increment = 0.2) does not converge while the changed case (iteration increment = 0.5) does after around 600 iterations. It seems like the base case reaches the same value as the changed case which is at convergence which means it could have just reached convergence; however, it is hard to tell with the negative slope of the base case graph between 600 and 700 iterations. Between around 350 to 550 iterations, the temperature difference between the base case and the changed case are slightly different with this location having the largest difference. However, the temperature difference is very small with the entire y-axis range only being 1 K. It can be concluded that the changed case reaches convergence faster for the vertex maximum of temperature than the base case does, but the differences are small.